

application sets respectively. In particular, employing a reduced set of templates for template selection leads to an average area reduction of over 27%.

Finally, Table 3 reports on the efficiency that is obtained when the reduced set of templates and the full set of templates are used for template selection. The efficiency is calculated as shown in (8), where *ESCS* is the Effective Software Cycle Savings which accounts for the hardware execution delay of the custom instructions p_i ($T_C(p_i)$), where $i = 1, \dots, n$, and is calculated as shown in (9). $T_C(p_i)$ is obtained by calculating the number of clusters in the critical path of the custom instructions. For example, T_C of the custom instruction in Figure 4(a) is 2. Note that we have assumed that the delay of each cluster is equivalent to two software clock cycle executions. This is a reasonable assumption as the FPGA logic can generally execute at a significantly higher clock frequency than a commercially available soft processor core which is implemented on the same fabric [55].

$$Efficiency = \frac{ESCS}{A_{RFU}^K} \quad (8)$$

$$ESCS = \sum_{i=1}^n F(p_i) \times (T_{sw}(p_i) - 2 \cdot T_C(p_i)) \quad (9)$$

Domain	Reduced Templates			Full Templates		
	ESCS	Area	Efficiency	ESCS	Area	Efficiency
Automotive-Industrial	14664576	45182539.6	0.325	14814576	64546485.15	0.230
Image	1538018	364687641.1	0.004	1559449	416324829.2	0.004
Network	102859967	64546485.15	1.594	129603704	80683106.43	1.606
Security	19207719	142002267.3	0.135	22467162	264640589.1	0.085
Telecomm	5823638	41955215.34	0.139	6873517	71001133.66	0.097
Generic	146359636	635782878.68	0.230	176180091	761648524.7	0.231

Table 3: Comparing the efficiency when reduced set and full set of templates are used for template selection

It can be observed in Table 3 that the efficiency of the proposed method is higher than the case when the full set of templates is used in a number of application domains (with comparable results in the remaining domains). In particular, the average efficiency gain when a reduced set of templates is used is over 25%.